RelayAI Router Explorer

Installation guide — run the router explorer on your own laptop, macOS or Windows.

Written for readers who have never used a terminal. Every command is given in full; you can copy and paste each one. Nothing here needs prior programming experience.

What you are installing

A small web app that shows the trained LLM router making one decision at a time: it reads your question, scores every model in the pool, adds the price, and picks one. It runs entirely on your own machine — nothing is sent anywhere.

It has **two halves that you start separately**, each in its own terminal window, and both must be running at the same time:

- the **backend** — the part that holds the trained network and does the thinking. It is written in **Python**.
- the **frontend** — the part you actually look at in your browser. It is built with **Node.js**.

That is why you install two different things below. Set aside about **30 minutes** the first time, most of it waiting for downloads.

> **You will need internet** for the installation and for the first start of the backend, which fetches a language model of about 87 MB. After that the app works offline. Budget about **1.5 GB of free disk space**. No special graphics card is needed.

1 Opening a terminal

A **terminal** is a window where you type commands instead of clicking. You will use it a lot below, so open one now and keep it handy.

MACOS

Press `Cmd + Space`, type **Terminal**, press `Enter`. A window with a text prompt appears.

To open a **second** one later, press `Cmd + N`.

WINDOWS

Press the **Windows key**, type **PowerShell**, click **Windows PowerShell**.

For a **second** one later, just repeat this.

To run a command: copy it, click into the terminal, paste, press `Enter`. Then **wait** until the prompt comes back before typing the next one.

> **Pasting.** On macOS use `Cmd + V`. In PowerShell use `Ctrl + V`, or simply right-click. Nothing appears while you type a password — that is normal.

2 Do you already have Python and Node?

Run these two lines. If each prints a version number, skip ahead to section 5.

MACOS

```
python3 --version
node --version
```

WINDOWS

```
python --version
node --version
```

If you instead see *command not found*, *not recognized*, or the Microsoft Store opens, that piece is missing. Install it in the next two sections.

Python must be **3.10 or newer**, Node **18 or newer**. If yours is older, install the newer version the same way.

3 Installing Python

MACOS

1. Go to python.org/downloads/macos.
2. Click the yellow **Download Python 3.13** button (any 3.10+ is fine). A `.pkg` file lands in your Downloads.
3. Double-click that file. Click **Continue** through the installer, agree to the licence, then **Install**. Enter your Mac password when asked.
4. Close the installer, **close your terminal window and open a new one**, then check:

```
python3 --version
```

WINDOWS

1. Go to python.org/downloads/windows.
2. Click **Download Python 3.13**, then open the `.exe` file that downloads.
3. **Before clicking anything else**, tick the box at the bottom of the first screen that says **"Add python.exe to PATH"**. This is the single most common thing to get wrong — without it, Windows will not find Python later.
4. Click **Install Now** and let it finish.
5. **Close PowerShell and open a new one**, then check:

```
python --version
```

> **If Windows opens the Microsoft Store** when you type `python`, the PATH box was not ticked. Re-run the installer, choose **Modify**, and make sure the PATH option is on. Then open a fresh PowerShell.

4 Installing Node.js

WINDOWS

1. Go to nodejs.org/en/download.
2. Choose the version marked **LTS** — it means long term support, the stable one. Do not take "Current".
3. Make sure the platform says **Windows** and the format is the **Windows Installer (.msi)**, then download it.
4. Open the downloaded `.msi` file. Click **Next**, accept the licence, and keep clicking **Next** — the defaults are correct. Node adds itself to PATH on its own, so there is no box to tick here.
5. On the screen offering **Tools for Native Modules**, leave it **unticked**. You do not need it, and it installs several gigabytes.
6. Click **Install**, approve the Windows prompt, then **Finish**.
7. **Close PowerShell and open a new one**, then check both:

```
node --version
npm --version
```

Two version numbers means you are done. `npm` comes with Node; you do not install it separately.

MACOS

1. Go to nodejs.org/en/download and choose the **LTS** version, platform **macOS**, format **.pkg**.
2. Open the downloaded file and click **Continue** and **Install** through the installer.

3. Open a new terminal window and check:

```
node --version
npm --version
```

5 Unpacking the zip

Find `RelayAI-Router-Explorer.zip` in your Downloads folder.

- MACOS double-click it. A folder appears next to it.
- WINDOWS right-click it, choose **Extract All…**, then **Extract**. Do not skip this — Windows will let you look inside a zip without unpacking it, but the app cannot run from there.

Everything below assumes the folder ended up in **Downloads**. If you moved it elsewhere, change the path in the first `cd` command and leave the rest exactly as written.

> **What cd means.** It stands for *change directory* — it tells the terminal which folder to work in, the same way double-clicking a folder does in Finder or Explorer. Every session starts with one.

6 The requirements file

The backend needs nine Python packages. You do not hunt for them one by one: they are already listed for you in a plain text file inside the folder you just unpacked, at

```
app/backend/requirements.txt
```

There is nothing to download. It came with the zip. The `pip install -r` command in the next section reads that file and fetches everything in it in one go. This is what it contains:

```
# backend: serves relay_router over HTTP
fastapi>=0.110
uvicorn[standard]>=0.27
pydantic>=2.6
# relay_router itself
numpy>=1.24
pandas>=2.0
torch>=2.1
scikit-learn>=1.3
sentence-transformers>=2.5
```

The `-r` in the command is short for *requirements*. The `>=` means “this version or newer”. `torch` is PyTorch and is by far the largest, which is why that step takes a few minutes.

7 MACOS Starting the app

Terminal 1 — the backend

1. **Go to the folder.**

```
cd ~/Downloads/RelayAI-Router-Explorer
```

2. **Make a private space for the packages**, so they cannot disturb anything else on your Mac. This creates a hidden `.venv` folder and switches into it.

```
python3 -m venv .venv
source .venv/bin/activate
```

Your prompt now begins with `(.venv)`. That is how you know it worked.

3. **Install the nine packages** from the requirements file. Several minutes the first time. Lots of scrolling text is normal.

```
pip install --upgrade pip
pip install -r app/backend/requirements.txt
```

4. **Start the backend.**

```
python -m uvicorn app.backend.main:app --port 8000
```

Wait for the line `Application startup complete`. The first time, it downloads the language model before that appears, so give it a minute. **Leave this window open and running.** It looks like it has frozen; it has not.

Terminal 2 — the frontend

Open a **second** terminal window with `Cmd + N`. Do not close or reuse the first one.

1. **Go to the frontend folder.**

```
cd ~/Downloads/RelayAI-Router-Explorer/app/frontend
```

2. **Install the web packages.** About 120 of them, usually a few seconds.

```
npm install
```

3. **Start it.**

```
npm run dev
```

4. **Open the app** in your browser at http://localhost:5173.

8 WINDOWS Starting the app

PowerShell 1 — the backend

1. **Go to the folder.**

```
cd $HOME\Downloads\RelayAI-Router-Explorer
```

2. **Make a private space for the packages.**

```
python -m venv .venv
.\.venv\Scripts\Activate.ps1
```

If that second line is refused with a message about scripts being disabled, run the line below and then try activating again. It applies only to this window.

```
Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass
```

Your prompt now begins with `(.venv)`.

3. **Install the nine packages** from the requirements file. Several minutes the first time.

```
python -m pip install --upgrade pip
pip install -r app\backend\requirements.txt
```

4. **Start the backend.**

```
python -m uvicorn app.backend.main:app --port 8000
```

Wait for `Application startup complete`. **Leave this window open and running.**

PowerShell 2 — the frontend

Open a **second** PowerShell window. Do not close the first one.

1. **Go to the frontend folder.**

```
cd $HOME\Downloads\RelayAI-Router-Explorer\app\frontend
```

2. **Install the web packages.**

```
npm install
```

3. **Start it.**

```
npm run dev
```

4. **Open the app** at http://localhost:5173.

9 Stopping, and starting again another day

To stop: click each terminal window and press `Ctrl + C`. Do this in both. On macOS you can then type `deactivate` to step out of the private package space.

Next time you do not install anything again. Only these:

MACOS

WINDOWS

```
# terminal 1
cd ~/Downloads/RelayAI-Router-Explorer
source .venv/bin/activate
python -m uvicorn app.backend.main:app --port 8000

# terminal 2
cd ~/Downloads/RelayAI-Router-Explorer/app/frontend
npm run dev
```

```
# powershell 1
cd $HOME\Downloads\RelayAI-Router-Explorer
.\.venv\Scripts\Activate.ps1
python -m uvicorn app.backend.main:app --port 8000

# powershell 2
cd $HOME\Downloads\RelayAI-Router-
Explorer\app\frontend
npm run dev
```

10 If something goes wrong

WHAT YOU SEE	WHAT TO DO
A red banner: *cannot reach the router backend at http://127.0.0.1:8000*	The backend is not running yet, or still starting. Go to terminal 1 and wait for `Application startup complete`. Always start the backend before the frontend, then reload the browser.
`python: command not found` / *not recognized*	Python is not installed, or the terminal was already open when you installed it. Close every terminal window, open a new one, try again. On macOS type `python3`, on Windows `python`.
Typing `python` opens the Microsoft Store	The **"Add python.exe to PATH"** box was not ticked. Re-run the Python installer, choose **Modify**, enable it, then open a fresh PowerShell.
running scripts is disabled on this system	Run `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass` in that same window, then run the activate line again.
`address already in use` on port 8000	Another program holds that port. Use a different one, and tell the frontend where to look. MACOS `python -m uvicorn app.backend.main:app --port 8010`, then in terminal 2 `VITE_API_BASE=http://127.0.0.1:8010 npm run dev` WINDOWS same first line, then `$env:VITE_API_BASE="http://127.0.0.1:8010"; npm run dev`
Port 5173 already in use	The frontend picks the next free port by itself and prints it. Open the address it prints rather than 5173.
The `pip install` seems stuck	PyTorch is several hundred megabytes. Give it a few minutes on a normal connection before assuming anything is wrong.
A warning mentioning `HF_TOKEN` or unauthenticated requests	Harmless. It only notes you are not signed in while the language model downloads. Ignore it.
The terminal looks frozen after starting the backend	That is correct. A running server keeps its window busy. Leave it alone and use your second window.
The page loads but all the panels are empty	The frontend started before the backend was ready. Reload the browser once the backend says `Application startup complete`.

11 What is in the folder

PATH	WHAT IT IS
`Installation Guide.pdf`	This document.
`app/backend/`	The Python half. `requirements.txt` is the package list from section 6.
`app/frontend/`	The part you see in the browser: the three panels.
`relay_router.py`	The router itself. Every number the app shows comes from here.
`router_bundle/`	The trained network and the measured tables it reads. Nothing trains when you run the app.